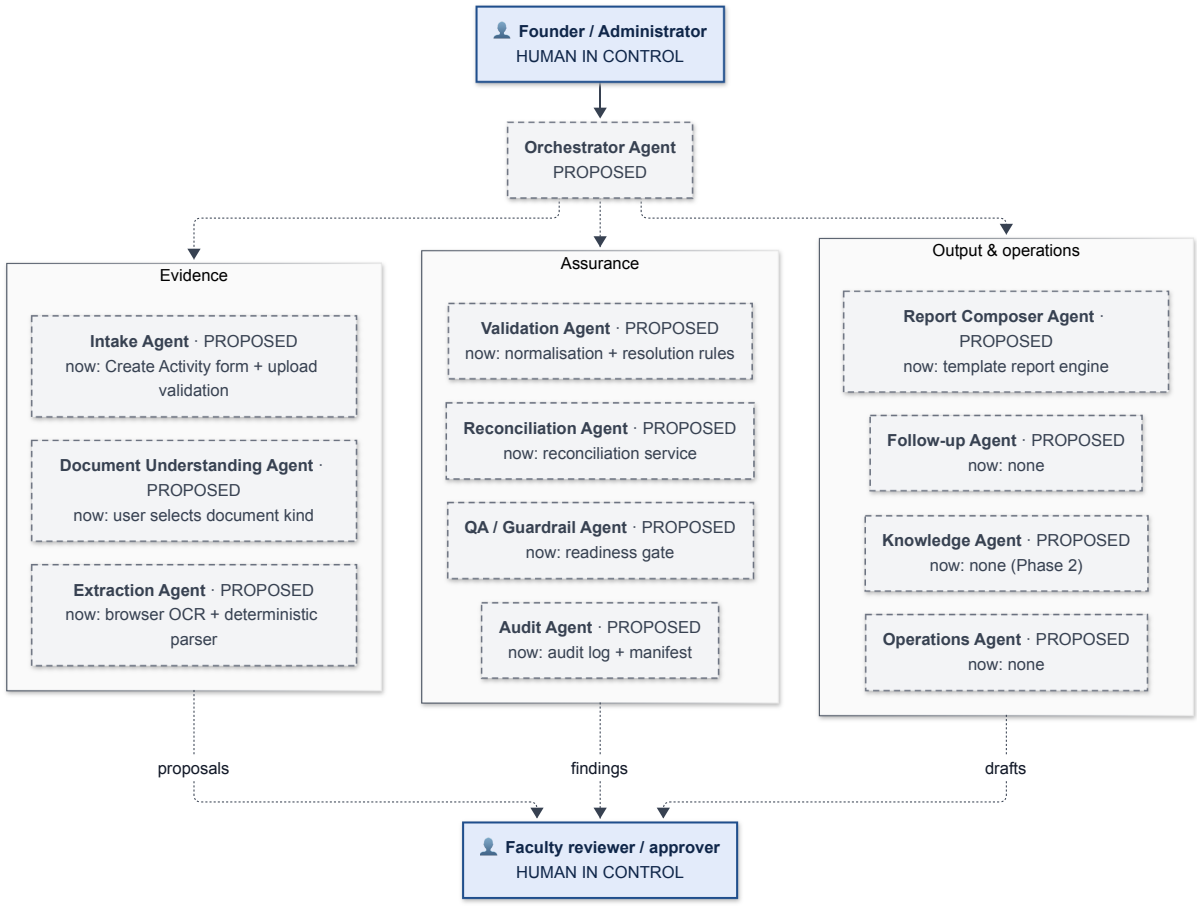
College Operations Copilot — Agent Handbook

The proposed "AI workforce", in plain language · Version 1.0 · 29 September 2026 · software at commit `7119b13`

> **Read this first.** Every agent in this handbook is a PROPOSED AGENT. None of them exists as an autonomous agent today. Each is paired with the CURRENT SOFTWARE SERVICE that does the equivalent job in the implemented system: deterministic code, tested, with no AI model involved. The one AI capability that exists today is the OPTIONAL LLM-assisted extraction adapter, which is off by default. HUMAN marks people.

How the workforce is organised



Legend Proposed / future — not built · Human control point · → arrows show data / control direction; dotted = optional or proposed

Agent map: proposed agents (grey dashed), grouped by responsibility, with the people who stay in control (blue). Each box names the current software counterpart.

Six rules every agent follows

These are design rules for any future version. Rules 3 and 4 are already enforced by the current code.

1. **Agents propose; people decide.** An agent may create candidate values, draft text or message drafts. It never approves, archives, reopens or publishes anything.
2. **Same door as people.** Agents act only through the existing service functions, so validation, access control and audit logging apply to them exactly as they do to a logged-in user. Each action is logged with the agent's name, model and prompt version.

3. **No quote, no fact.** A factual value must cite a document page and a verbatim quote from it. The current `groundLlmValue` check discards anything else. ENFORCED TODAY
4. **The gate is code, not an agent.** The readiness gate blocks approval while conflicts, missing required fields, unconfirmed low-confidence values, count mismatches or stale reports remain. No agent can call or skip it. ENFORCED TODAY
5. **Escalate, don't guess.** Missing or contradictory evidence goes to a person with an explanation. The value stays empty.
6. **Nothing leaves the building without a person.** Emails and messages stay drafts until a human sends them.

The people in control

1. Founder / Administrator HUMAN

Job	Owns the system: configures templates, creates user accounts, sets agent permissions (future), reviews the audit trail.
Input	Institutional policies, template wording, user list, audit reports.
Output	Template versions, accounts, policy decisions, reopened activities with a stated reason.
Tools	Templates editor, Settings → Users, Audit Log (all implemented).
Authority	Highest. Only role that can edit templates, manage users or reopen an approved activity.
Restrictions	Every action is audited; reopening requires a written reason.
Escalation path	Institution leadership (HoD / IQAC / Principal) for policy questions.

1a. Faculty reviewer / approver HUMAN

Job	Owns each activity: uploads evidence, reviews every extracted value, resolves conflicts, verifies attendance, approves.
Input	Documents, the Review / Attendance / Reports screens.
Output	Choices, confirmations, edits (original kept), acknowledgements, approval.
Authority	The only person who can approve their activity (administrators can as well). Approval records `approved_by` and `approved_at`.
Restrictions	Cannot approve while any blocking issue exists. After approval the activity is read-only.
Escalation path	Administrator (for example, to reopen).

2. Orchestrator Agent PROPOSED AGENT

Job	Plans the work for an activity and hands tasks to the other agents in the right order.
Input	Activity state, list of uploaded documents, open issues.
Output	A task plan and a progress summary for the faculty member.
Tools	Read activity state; invoke other agents.
Authority	Coordination only.
Restrictions	Cannot write field values, cannot call the approval service, cannot skip validation, reconciliation or review.
Escalation path	Stops and reports to the faculty member when an agent fails or evidence conflicts.
Current counterpart	CURRENT SOFTWARE SERVICE The workflow tabs and `refreshStatus` status logic. The person drives the order today.

Evidence agents

3. Intake Agent PROPOSED AGENT

Job	Collects what is needed for the chosen template: asks for missing details and documents, and checks that nothing obvious is forgotten (for example, the attendance sheet).
Input	Faculty answers, uploaded files, the template's required fields.
Output	A draft activity and a checklist of missing evidence.
Tools	Read the template; create a draft activity; upload through the normal upload validation.
Authority	Can create a draft and attach files the user provides.
Restrictions	Never fills a field the user did not state; never uploads files from outside the conversation.
Escalation path	Asks the faculty member.
Current counterpart	CURRENT SOFTWARE SERVICE The Create Activity form (all fields optional, nothing pre-filled) plus upload validation.

4. Document Understanding Agent PROPOSED AGENT

Job	Works out what each document, or each page of a compiled scan, is: notice, poster, report, attendance sheet, photograph.
Input	Uploaded PDF or image.
Output	Page-kind proposals with a reason for each.
Tools	Read the file and its page text.
Authority	Proposals only.
Restrictions	Cannot reclassify a document without the user accepting; never reads facts from photographs.
Escalation path	Unclear pages are flagged for the user to label.
Current counterpart	CURRENT SOFTWARE SERVICE The user picks the document kind at upload. Per-page classification is not built.

5. Extraction Agent (OCR / extraction) PROPOSED AGENT

Job	Reads text (text layer or OCR) and proposes field values with evidence.
Input	Document pages.
Output	Field candidates, each with page, verbatim snippet, method, rule and confidence.
Tools	pdf.js, tesseract.js, deterministic parser, optionally an LLM with grounding.
Authority	Writes candidates only.
Restrictions	Every value needs a quote from the cited page. OCR values under 0.8 confidence must be confirmed. Handwritten attendance is always drafted as unverified.
Escalation path	Marks the extraction failed or incomplete and tells the user which pages could not be read.
Current counterpart	CURRENT SOFTWARE SERVICE Browser extraction + deterministic parser. OPTIONAL LLM-assisted extraction adapter (off by default).

Assurance agents

6. Validation Agent PROPOSED AGENT

Job	Explains each open issue in plain language and points the reviewer to the exact evidence.
Input	Field states, candidates, readiness issues.
Output	Issue explanations (“the poster shows 31 Oct, the notice and report show 6 Nov — the poster may show the commemorated day”).
Tools	Read-only access to fields, candidates and page text.
Authority	Comments only.
Restrictions	Cannot resolve a conflict, confirm a value or change a status.
Escalation path	The reviewer.
Current counterpart	CURRENT SOFTWARE SERVICE Type normalisation, resolution rules (`fields/resolve.ts`) and the readiness gate (`readiness.ts`), with issue messages in the UI.

7. Reconciliation Agent PROPOSED AGENT

Job	Helps check attendance row by row against the original sheet, and explains count mismatches.
Input	Attendance sheet pages, drafted rows, stated male / female / total counts.
Output	Suggested row corrections and a mismatch explanation.
Tools	Read files and rows; propose row edits.
Authority	Proposals only.
Restrictions	Never marks a row verified. Never infers gender from a name. Never changes stated counts.
Escalation path	The reviewer verifies rows or records a written acknowledgement.
Current counterpart	CURRENT SOFTWARE SERVICE Reconciliation service (PASS / FAIL / UNVERIFIED / INSUFFICIENT_DATA) and the attendance review screen.

8. QA / Guardrail Agent (quality & compliance) PROPOSED AGENT

Job	Checks every sentence of a draft report against the evidence before the faculty member sees it.
Input	Draft text, fields, candidates and snippets.
Output	Pass, or a list of unsupported statements.
Tools	Read-only.
Authority	Can block a draft from going to review.
Restrictions	Cannot edit or approve.
Escalation path	Report Composer (redraft) and then the reviewer.
Current counterpart	CURRENT SOFTWARE SERVICE Reports are built only from template text plus field values, so there is no free text to check. Missing values print “Not provided”.

9. Audit Agent PROPOSED AGENT

Job	Summarises the history of an activity and flags unusual patterns (for example, many edits after reports were generated).
Input	Audit log, provenance manifest.
Output	A review summary for the administrator.
Tools	Read-only.
Authority	Reporting only.
Restrictions	Cannot edit or delete audit records (the log is append-only).
Escalation path	Administrator.
Current counterpart	CURRENT SOFTWARE SERVICE Append-only `audit_logs`, the History tab, the Audit Log page and the JSON provenance manifest.

Output and operations agents

10. Report Composer Agent PROPOSED AGENT

Job	Drafts more natural wording for narrative sections, using only reviewed field values.
Input	Reviewed fields and the template.
Output	Draft narrative text, marked as a draft.
Tools	Read fields; write draft text.
Authority	Drafts only.
Restrictions	Cannot add facts, names, dates, counts, objectives or outcomes that are not in the fields. Every draft passes the QA / Guardrail Agent. The faculty member accepts or edits it.
Escalation path	Leaves the section as “Not provided” when evidence is missing.
Current counterpart	CURRENT SOFTWARE SERVICE Deterministic report engine (`reports/build.ts`). The detailed-report draft only combines existing fields and says which ones it used.

11. Knowledge Agent PROPOSED AGENT — Phase 2

Job	Answers questions such as “When are examinations?” from the academic calendar, timetable and notices.
Input	A question plus approved institutional documents.
Output	An answer with the source document and page, or “Not found in the provided documents”.
Tools	Read-only search over stored page text.
Authority	Read-only.
Restrictions	Only answers from uploaded, approved sources and always cites them.
Escalation path	Directs the user to the responsible office.
Current counterpart	None. The page-text storage it would search (`source_extractions`) already exists.

12. Follow-up Agent PROPOSED AGENT

Job	Drafts reminders for missing evidence, unverified attendance or pending approvals.
Input	Open issues and activity status.
Output	Message drafts.
Tools	Draft messages only.
Authority	None to send.
Restrictions	A person reviews and sends every message.
Escalation path	Activity owner.
Current counterpart	None. The dashboard shows counts by status.

13. Operations Agent PROPOSED AGENT — Phase 5

Job	Tracks recurring operational tasks across departments and produces status digests.
Input	Task registers and activity statuses.
Output	A digest for owners.
Tools	Read-only; drafts.
Authority	Reporting only.
Restrictions	No changes to records.
Escalation path	Department owners.
Current counterpart	None.

What exists today vs what is proposed

Capability	Today	Proposed agent
Collect details and documents	CURRENT Create Activity form, upload validation	Intake Agent
Identify document types	CURRENT User selects the kind per upload	Document Understanding Agent
Read text and propose values	CURRENT Browser OCR + deterministic parser · OPTIONAL grounded LLM adapter	Extraction Agent
Detect conflicts and missing data	CURRENT Resolution rules + readiness gate	Validation Agent (explains issues)
Check attendance and counts	CURRENT Reconciliation service + review screen	Reconciliation Agent
Write reports	CURRENT Deterministic template engine	Report Composer Agent + QA / Guardrail Agent
Keep an audit trail	CURRENT Append-only log, manifest	Audit Agent (summaries)
Coordinate the steps	CURRENT Workflow tabs; the person drives	Orchestrator Agent
Answer questions, remind, run operations	Not built	Knowledge · Follow-up · Operations Agents
Approve	HUMAN only, always	**No agent — by design**

Why agents are introduced gradually. The trust model is already enforced in code: provenance, grounding, the gate and human approval. Each proposed agent can be added behind it one at a time, with its permissions limited to “propose”. Adding an agent therefore does not require re-proving that reports stay free of unsupported facts.

Originality and scope. This handbook describes a proposed agentic operating model purpose-built for college documentation operations. It does not claim that individual technologies or concepts are globally unprecedented. A novelty claim would require separate prior-art and competitive research.